

DDAVP REPLACES AVP MONITOR Na CLOSELY

DDAVP ESCAPE

nephrogenic may resist to High Dose

FOUR DDAVP ROUTES

- IV SC:** 1 TO 2 mcg, ONSET 15 MIN
- INTRANASAL:** 10 TO 20 mcg BID TID
- ORAL:** 100 TO 400 mcg, ONSET 60 MIN, BID TID
- ORAL MELT**

one step at a time every 1-2 hours

PARTIAL AVP RESISTANCE COMBO
▲ NEPHROTOXICITY MONITOR Cr

PARTIAL RESISTANCE COMBO
Low-Sodium Diet

COMBINE - Additive effect
AVP RESISTANCE TREATMENT complete

HYPONATREMIA RISK

JUMPING BACKWARD

CLOCK ANALOGY: Intermittent delay equals transient aquaresis prevents hyponatremia.

HYPONATREMIA RISK CHART:

Severity	Risk
MILD	25% (131 to 134)
SEVERE	15% (less than 130)

low dose remain high AVP levels + increased fluid intake make decrease... leads to extra fluid body absorb... more fluid in urinary system (concentrated)

LITHIUM NDI

DANGER: VOLUME CONTRACTION CAN CAUSE LITHIUM TOXICITY MONITOR LEVELS

PRIMARY POLYDIPSIA SUPPORTIVE

- FLUID RESTRICTION COVERED
- GLP-1 ANALOGUES
- ICE CHIPS
- HARD CANDY

SEE ADIPSIC DIET FOR ADIPSIC PROTOCOL

1. DDAVP replaces AVP — monitor Na closely

WHAT YOU SEE

Banner 'DDAVP REPLACES AVP MONITOR Na CLOSELY'.

WHAT IT ENCODES

Desmopressin (DDAVP) is the treatment for central DI (AVP deficiency), replacing the missing hormone via V2 receptors. The dominant safety issue is iatrogenic hyponatremia, so serum sodium must be monitored.

BOARD TRAP

DDAVP only helps central DI — using it for nephrogenic DI fails, and over-dosing any patient causes dilutional hyponatremia.

2. Four DDAVP routes

WHAT YOU SEE

Left panel 'FOUR DDAVP ROUTES': IV/SC, intranasal, oral, oral melt.

WHAT IT ENCODES

DDAVP is given IV/SC (1–2 mcg, onset ~15 min), intranasal (10–20 mcg BID), oral (100–400 mcg BID-TID), or oral melt/sublingual. Bioavailability differs greatly, so doses are NOT interchangeable mg-for-mg.

BOARD TRAP

Switching routes without re-titrating is dangerous — oral DDAVP needs far higher mcg than IV; copy-pasting the dose causes under- or over-treatment.

3. IV/SC route

WHAT YOU SEE

'IV SC 1 to 2 mcg, onset 15 min'.

WHAT IT ENCODES

Parenteral DDAVP (1–2 mcg SC/IV) has the fastest onset (~15 min) and highest potency — used acutely (post-op DI, unable to take orals). Requires careful fluid balance.

BOARD TRAP

Parenteral DDAVP is the most potent route — applying an oral mcg dose IV would massively overdose the patient.

4. Intranasal route

WHAT YOU SEE

'INTRANASAL 10 to 20 mcg BID'.

WHAT IT ENCODES

Intranasal DDAVP (10–20 mcg BID) is convenient for chronic central DI but absorption is erratic with rhinitis/mucosal disease. Titrate to allow a daily 'breakthrough' to avoid water retention.

BOARD TRAP

Nasal congestion alters intranasal absorption unpredictably — blaming 'treatment failure' may just be impaired mucosal uptake.

5. Oral / oral melt

WHAT YOU SEE

'ORAL 100 to 400 mcg... ORAL MELT'.

WHAT IT ENCODES

Oral DDAVP (100–400 mcg) and sublingual melt have low bioavailability, requiring much larger mcg doses than nasal/parenteral. Preferred for stable outpatient central DI.

BOARD TRAP

Oral DDAVP mcg numbers are large precisely because absorption is poor — never equate them to the tiny IV/SC dose.

6. DDAVP escape — intermittent delay = transient aquaresis

WHAT YOU SEE

Clock 'DDAVP ESCAPE — jumping backward'; 'intermittent delay equals transient aquaresis prevents hyponatremia'.

WHAT IT ENCODES

Intentionally allowing the DDAVP effect to wane between doses ('DDAVP escape') lets the patient excrete free water periodically, preventing chronic water retention and hyponatremia. Build a daily breakthrough into dosing.

BOARD TRAP

Round-the-clock DDAVP with no gap removes the escape window and steadily lowers sodium — the most common complication of DDAVP therapy is hyponatremia.

7. Hyponatremia risk — most common complication

WHAT YOU SEE

Center 'MOST COMMON COMPLICATION OF DDAVP THERAPY'; pie 'hyponatremia risk'.

WHAT IT ENCODES

Hyponatremia is the chief adverse effect of DDAVP, from continued water intake plus antidiuresis. Risk rises with excess fluids and missed escape windows; severe (<130) hyponatremia can cause seizures.

BOARD TRAP

Symptomatic hyponatremia, not hypernatremia, is the DDAVP danger — counsel fluid moderation and check Na, especially when starting/escalating.

8. Hyponatremia severity — ~25% any, ~15% severe <130

WHAT YOU SEE

Pie chart '25 percent... 15 percent severe less 130'.

WHAT IT ENCODES

Roughly a quarter of DDAVP-treated patients develop some hyponatremia and ~15% severe (<130 mmol/L). Drives the practice of periodic monitoring and built-in aquaresis windows.

BOARD TRAP

Even 'stable' DDAVP patients need periodic Na checks — a significant minority drift to severe hyponatremia silently.

9. Lithium NDI — amiloride blocks ENaC entry

WHAT YOU SEE

Right panel 'LITHIUM NDI — amiloride blocks lithium entry through ENaC'.

WHAT IT ENCODES

For lithium-induced nephrogenic DI, amiloride blocks the ENaC channel through which lithium enters principal cells, limiting AQP2 downregulation. HCTZ reduces volume and paradoxically reduces urine output.

BOARD TRAP

Amiloride is the targeted agent for lithium NDI; adding a thiazide raises lithium levels (volume contraction), so monitor lithium for toxicity.

10. HCTZ + volume-contraction caution

WHAT YOU SEE

Right panel 'HCTZ... DANGER volume contraction can cause lithium toxicity, monitor levels'.

WHAT IT ENCODES

Thiazides treat nephrogenic DI by inducing mild volume contraction that enhances proximal Na/water reabsorption, cutting distal water delivery and urine output. In lithium NDI they can raise lithium levels.

BOARD TRAP

Thiazide for nephrogenic DI works by VOLUME contraction — but in lithium patients that same contraction precipitates lithium toxicity, so check levels.

11. Nephrogenic DI treatment combo

WHAT YOU SEE

Bottom-left 'AVP RESISTANCE TREATMENT': low-sodium diet, thiazide, amiloride, NSAID.

WHAT IT ENCODES

Nephrogenic DI is managed with a low-solute (low-Na, low-protein) diet, thiazide diuretic, amiloride (esp. lithium), and an NSAID (indomethacin) which blunts prostaglandin antagonism of AVP. DDAVP is ineffective.

BOARD TRAP

NSAIDs reduce urine output in nephrogenic DI by removing prostaglandin inhibition of water reabsorption — but watch renal function; and remember DDAVP won't work here.

12. Partial resistance combo / monitor

WHAT YOU SEE

Left 'PARTIAL AVP RESISTANCE COMBO' / 'PARTIAL RESISTANCE COMBO'.

WHAT IT ENCODES

Partial nephrogenic DI may respond to high-dose DDAVP combined with thiazide/amiloride and solute restriction. Monitor urine output and creatinine to titrate the combination.

BOARD TRAP

Partial resistance can show some DDAVP response, blurring central vs nephrogenic — copeptin testing clarifies before committing to a regimen.

13. Primary polydipsia — fluid restriction supportive

WHAT YOU SEE

Bottom-right 'PRIMARY POLYDIPSIA SUPPORTIVE': fluid restriction, AVP analogues, ice chips, hard candy.

WHAT IT ENCODES

Primary polydipsia is treated by reducing water intake (behavioral fluid restriction) plus symptom aids (ice chips, hard candy for dry mouth). DDAVP is generally avoided due to hyponatremia risk.

BOARD TRAP

Giving DDAVP/AVP analogues to a primary-polydipsia patient who keeps drinking precipitates dangerous hyponatremia — restrict water instead.

14. See adipsic ward for adynamic thirst

WHAT YOU SEE

Bottom-right 'see adipsic ward for adynamic thirst'.

WHAT IT ENCODES

Adipsic (adynamic-thirst) DI patients lack the thirst safeguard, so they cannot self-correct rising sodium — they need scheduled fixed water intake plus DDAVP and frequent Na monitoring.

BOARD TRAP

Adipsic DI is the highest-risk subtype: without thirst, both hyper- and hyponatremia occur easily — fixed fluid prescriptions, not 'drink to thirst', are required.
